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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester B.Tech (CBCS) Degree Examinations

December 2024

EE1001-2 – BASIC ELECTRICAL ENGINEERING
(For ACT, BT, EC, EE, VLSI)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. Which of the following according to KVL must be zero?
A) Algebraic sum of currents in closed-loop B) Algebraic sum of power in closed-loop
C) Algebraic sum of losses in closed-loop D) Algebraic sum of voltages in closed-loop
2. Inductor does not allow the sudden change of
A) Current B) Voltage
C) Power D) Resistance
3. For Sine waveform average value of one full cycle is
A) 1 B) 0
C) 1.11 D) 2.22
4. In a series R, L circuit, voltage across resistor and inductor are 3 V and 4 V respectively, then what is the applied voltage
A) 7V B) 4V
C) 5V D) 3V
5. Average power taken by the pure capacitor is
A) Minimum B) Zero
C) Maximum D) Infinite
6. Kirchhoff's Law is applicable to
A) Passive network only B) A.C. Network only
C) D.C. network only D) Both A.C and D.C circuits
7. In a RC series circuit total voltage is 10V and voltage across resistor is 6V, then what is voltage across capacitor
A) 4V B) 8V
C) 16V D) 10V
8. In RLC series circuit, if the voltage across the capacitor is greater than the voltage across inductor, then power factor of the network is
A) Lagging B) Unity
C) Leading D) Zero
9. Self-induced emf is produced when:
A) A magnet moves near a coil B) The current in a nearby coil changes
C) A coil moves in a magnetic field D) The current in a coil changes

10. The principle of operation of a DC motor is based on
 A) Lenz's Law B) Ohm's Law
 C) Faraday's Law of Electromagnetic Induction D) Kirchhoff's Law
11. A DC shunt motor has
 A) High starting torque B) Approximately constant speed
 C) Low starting torque D) Poor speed regulation
12. A starter is used in an induction motor to
 A) Limit the starting current B) Increase the starting torque
 C) Improve the power factor D) Reduce the speed
13. A single-phase induction motor
 A) Is self-starting B) Is not self-starting
 C) Has high starting torque D) Has high efficiency
14. A transformer that has a higher secondary voltage than its primary voltage is called a
 A) Step-down transformer B) Isolation transformer
 C) Step-up transformer D) Autotransformer
15. The principle of operation of a transformer is based on
 A) Kirchhoff's Law B) Lenz's Law
 C) Mutual induction D) Self-induction
16. A transformer has an output power of 1000W and losses of 50W. Its efficiency is
 A) 95% B) 85%
 C) 90% D) 80%
17. What does SMPS stand for?
 A) Switched Mode Power System B) Smart Mode Power Solution
 C) Switched Mode Power Supply D) Simple Mode Power Source
18. What is the primary function of an MCB?
 A) To supply power to a circuit B) To protect a circuit from overcurrent
 C) To control the speed of a motor D) To regulate voltage levels
19. Boost converter is a _____ converter
 A) Step down - Step up B) Step Up - Step down
 C) Step down D) Step up
20. What is necessary to provide a path to the leakage current to ground and protect personnel from the danger of shock.
 A) Fuse B) Circuit Breaker
 C) Earthing D) Insulator

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

1. a) Define the terms with respect to alternating quantity
 i. Form factor
 ii. Peak factor

Marks BT* CO* PO*

04 L*1 1 1,2

- b) Using mesh analysis, find the voltage drop across the 2Ω resistor in Fig. 1b

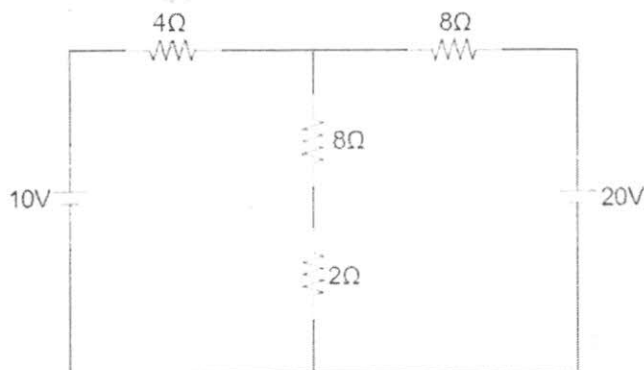


fig. 1b

06 L3 1 1,2

- c) Show that power consumed by a pure capacitor circuit is zero.
2. a) An alternating voltage has the equation $v = 325 \sin 314t$ Volts, what are the values of (a) r.m.s voltage (b) frequency
- b) Use nodal analysis to find the voltage across and current through 3Ω resistor in Fig. 2b

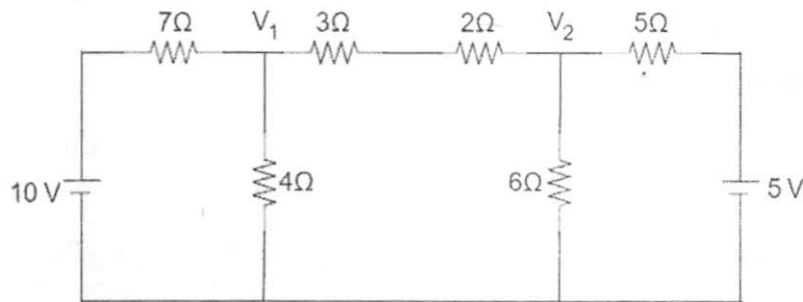


Fig 2b

- c) Derive the expression for three phase active power using two watt meter method of power measurement.
3. a) List the advantages of three phase connection over single phase connection
- b) A pure resistance of 20Ω is in series with an inductance of 30mH . The series combination is connected across a 230V , 50Hz supply. Find impedance, current and power factor of the circuit.
- c) Define average value of an alternating quantity. Derive an expression for average value of an alternating current.

Unit – II

4. a) Define self and mutually induced emf
- b) Prove that in three phase induction motor, stator magnetic field has constant magnitude and it rotates at synchronous speed
- c) A 200 kVA , 1-phase transformer has iron (core) losses of 1000 W and copper losses of 1500 W at full load. Calculate the efficiency of the transformer when it operates at 75% load and at a power factor of 0.8 lagging and half load with power factor 0.7 lagging.
5. a) Derive the EMF equation of a single phase transformer.
- b) What is an Autotransformer? List its advantages and applications.
- c) A four pole DC Motor is connected to a 400V DC supply and takes an armature current of 60A . The resistance of the armature is 0.4Ω . The armature is wave wound with 366 conductors and useful flux per pole is 0.025 Wb . Calculate the back emf and speed of the motor
6. a) With neat diagram explain constructional features of DC machines.
- b) A 2 pole 3 phase induction motor operates from a supply whose frequency is 50Hz . Calculate: i) The speed at which the magnetic field of the stator is rotating. ii) The speed of the rotor when the slip is 0.03 iii) Frequency of the rotor currents when the slip is 0.02 .
- c) Elaborate working principle of transformer.

Unit – III

7. a) List the Characteristics of materials used as fuse wires.
- b) With a neat block diagram, discuss the components of an electric vehicle.
- c) Explain the construction and working principle of PMSM.

06 L2 2 1,2

04 L3 1 1,2

06 L3 1 1,2

06 L2 2 1,2

04 L1 2 1,2

06 L3 2 1,2

06 L2 1 1,2

04 L1 3 1,2

06 L2 3 1,2

06 L3 4 1,2

06 L2 4 1,2

06 L1 4 1,2

04 L3 3 1,2

04 L1 3 1,2

06 L2 3 1,2

06 L2 4 1,2

04 L2 5 1,2

06 L1 5 1,2

06 L2 5 1,2

8. a) List the features of Brushless DC Motor which makes it ideal for electric vehicle applications.
- b) Discuss the principle of operation of UPS.
- c) What is earthing? With a neat diagram, explain pipe earthing.

04	L2	5
06	L2	5
06	L2	5

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
